CS-471: Machine Learning

Week 2

Instructor: Dr. Hashir Kiani

Classroom Etiquette and Participation Guidelines

- Treat everyone with respect and courtesy at all times
- Speak only when it is your turn and raise your hand before contributing to the discussion
- Do not interrupt others while they are speaking
- Join the class on time
- Keep mobile phones on silent during the lecture

LMS Enrolment Code

923870654

Lecture Outline

- Recap of Previous Lecture
- Linear Regression Example

Recap of Previous Lecture

- What is Machine Learning?
- What is the difference between AI, ML and DL?
- What are the three paradigms of Machine Learning?
- What is supervised learning? Regression and Classification?
- What is unsupervised learning? Reinforcement learning?
- What are some ML Applications? ML Risks?
- What is the objective of supervised learning? Training set?
- What is a linear regression model?
- What is hypothesis class? Loss function? Optimization Algorithm?

Recap of Previous Lecture

- What is score? Feature extractor? Feature vector? Weight vector?
- What is residual?
- What is the loss function used in linear regression?
- What is the gradient of this loss?
- What is the objective of optimization algorithm?
- How does gradient descent find the optimum weights?
- What is the formula for gradient descent?
- What are the possible problems with gradient descent?
- What is stochastic gradient descent?

Choosing Step Size

$$\boldsymbol{w}_{new} = \boldsymbol{w}_{old} - \alpha \nabla_{\boldsymbol{w}} TrainLoss(\boldsymbol{w})$$

- Could be constant
- Could be decreasing $(1/sqrt(number\ of\ updates\ made\ so\ far)$ Initial 1

Linear Regression Example

- Training data:

x	y
$[1,0]$	2
$[1,0]$	4
$[0,1]$	1

Questions?